Publication & Venue	Related Papers
IEEE VIS & TVCG	134
EuroVis & CGF	27
IEEE PacificVis	18
Other(CHI, JoV, etc)	33
Total	212

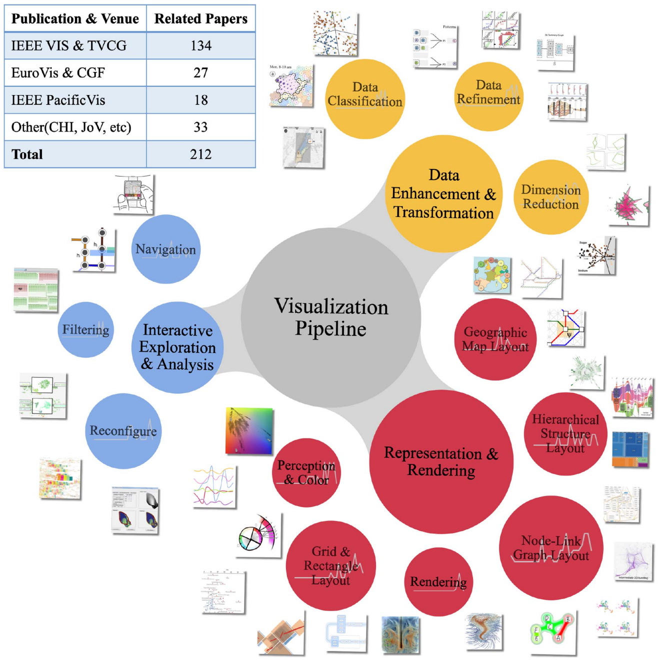

Fig. 1. Taxonomy of visualization pipeline. Each major category has corresponding minor categories. The bubble size encodes paper numbers. Each minor category is surrounded by selected literature. The sparkline in the bubble shows the temporal distribution of papers published in each minor category. The left-upper of the figure is a list of sources of paper, we use IEEE Xplore, ACM Digital Library, Google Scholar, and Wiley Online Library for literature retrieval.

Although this classification allows us to accurately obtain the application of various types of mathematical optimization in visualization, it cannot facilitate an in-depth exploration of how mathematical optimization is utilized in specific visualization forms or each phase of the visualization pipeline. Therefore, we adopt the categorization of visualization pipeline complemented by mathematical optimization and categorize papers in these two aspects. In each major category, we classify and summarize mathematical optimization and solution methods they used.

Visualization Pipeline Our taxonomy about visualization pipeline combines the following principles. First, we refer to the transition about VIS area model which began from 2020. The VIS area model contains six areas including *Theoretical & Empirical*, *Applications*, *Systems & Rendering*, *Representations & Interaction*, *Data Transformations*, and *Analytics & Decisions*. However, certain areas such as *Theoretical & Empirical*, aim to establish the foundation of VIS as a scientific subject and evaluate specific VIS techniques. Thus, there are fewer papers related to mathematical optimization in these areas. Second, a complete pipeline has already been established with the development of visualization research over the past few decades. McNabb and Laramee [6] summarize a basic pipeline in information visualization including data enhancement & transformation, visual mapping & structure, exploration & rendering, interactive analysis, and perception. Nevertheless, interactive analysis and perception have scarcer papers since they pay more attention to interaction techniques, analysis, and evaluation for data visualization which relate to mathematical optimization methods less.

In our topics of interest, interactive analysis and perception have relatively fewer papers related to mathematical optimization than the other three pipeline phases. Many papers use mathematical optimization models in scientific visualization, most of them center on the rendering phase or enhance data for rendering. Based on the above principles and considerations, we summarize the visualization pipeline into three phases in Fig. 1.

- *Data Enhancement & Transformation (DET)* address the issue of data processing before visual encoding. It is used to prepare data before visualization. This part includes *Dimension Reduction*, *Data Classification*, and *Data Refinement*. We describe them separately in Section III.
- *Representation & Rendering (RR)* has a strong emphasis on visual mapping and rendering, as well as encoding data structure to visual expression. Research on perception related to optimization can also be categorized into this phase because those papers describe quality of visual expression or layout. This part consists of *Grid & Rectangle Layout*, *Node-Link Graph Layout*, *Geographic Map Layout*, *Hierarchical Structure Layout*, *Perception & Color*, and *Rendering*. We discuss more details in Section IV.
- *Interactive Exploration & Analysis (IEA)* focuses on interaction, many papers related to mathematical optimization employ interactive techniques to fine-tune the visual layout by mathematical optimization. This part comprises *Navigation*, *Filtering*, and *Reconfiguration*. More details about this part are elaborated in Section V.

Mathematical Optimization Model We have already provided the basic concepts of mathematical optimization in the supplemental material, available online. We classify papers by basic mathematical optimization definitions to help readers understand their usage. Mathematical optimization can be categorized as four perspectives based on different conditions:

- *Constrained* or *Unconstrained* According to whether the mathematical optimization model has constraints, the mathematical optimization can be divided as constrained optimization or unconstrained optimization. For example, plenty of studies use energy function technique as an unconstrained optimization model to reduce visual confusion and improve visualization aesthetics. Constraints are ubiquitous in preserving the position relationship in visual layout problems.
- *Continuous*, *Discrete* or *Mixed* According to the range of variables, mathematical optimization can be divided into continuous, discrete, or mixed optimization. In visual layout, a common mathematical optimization problem is to compute and layout the continuous or discrete coordinates of points. A simple example of discrete optimization problem usually used in visual layout is to assign elements to designated areas.
- *Linear* or *Nonlinear* According to the form of the objective function and constraint, mathematical optimization can be divided into linear optimization and nonlinear optimization. In fact, due to the straightforward mathematical form,

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